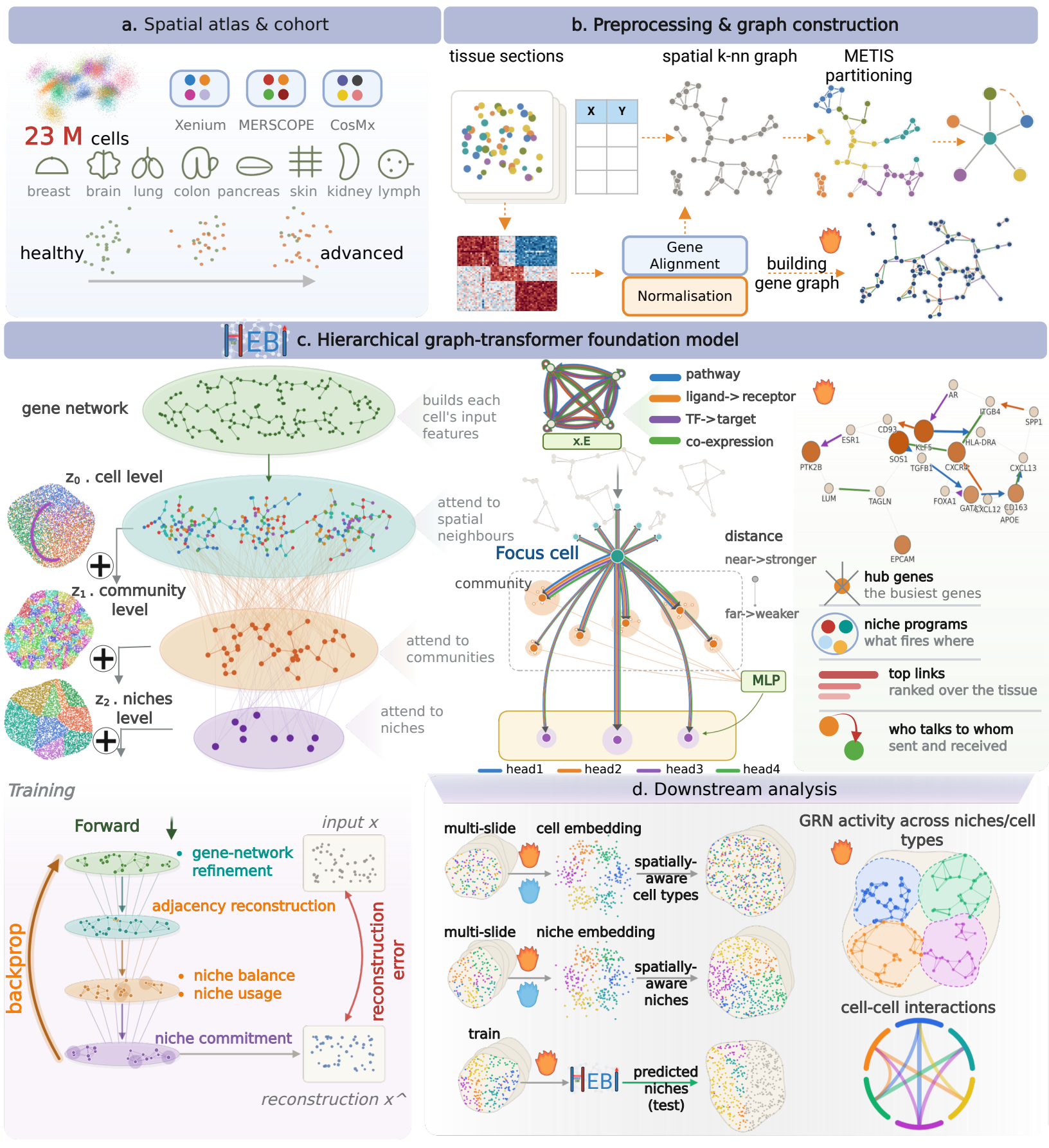

a. Spatial atlas & cohort
23 M cells
Xenium
MERSCOPE
CosMx
breast
brain
lung
colon
pancreas
skin
kidney
lymph
healthy
advanced
b. Preprocessing & graph construction
tissue sections
spatial k-nn graph
METIS partitioning
X
Y
Gene Alignment
Normalisation
building gene graph
c. Hierarchical graph-transformer foundation model
gene network
builds each cell's input features
z0 . cell level
attend to spatial neighbours
z1 . community level
attend to communities
z2 . niches level
attend to niches
x.E
pathway
ligand-> receptor
TF->target
co-expression
Focus cell
community
distance
near->stronger
far->weaker
MLP
head1
head2
head3
head4
AR
ITGB4
SPP1
CD93
ESR1
SOS1
KLF5
HLA-DRA
PTK2B
TGFB1
CXCR4
CXCL13
LUM
TAGLN
FOXA1
GATA3
CXCL12
CD163
APOE
EPCAM
hub genes
the busiest genes
niche programs
what fires where
top links
ranked over the tissue
who talks to whom
sent and received
Training
Forward
input x
gene-network refinement
adjacency reconstruction
niche balance
niche usage
niche commitment
backprop
reconstruction error
reconstruction x^
d. Downstream analysis
multi-slide
cell embedding
spatially-aware cell types
multi-slide
niche embedding
spatially-aware niches
train
predicted niches (test)
GRN activity across niches/cell types
cell-cell interactions